Building Knowledge Graphs

Building Knowledge Graphs mind map

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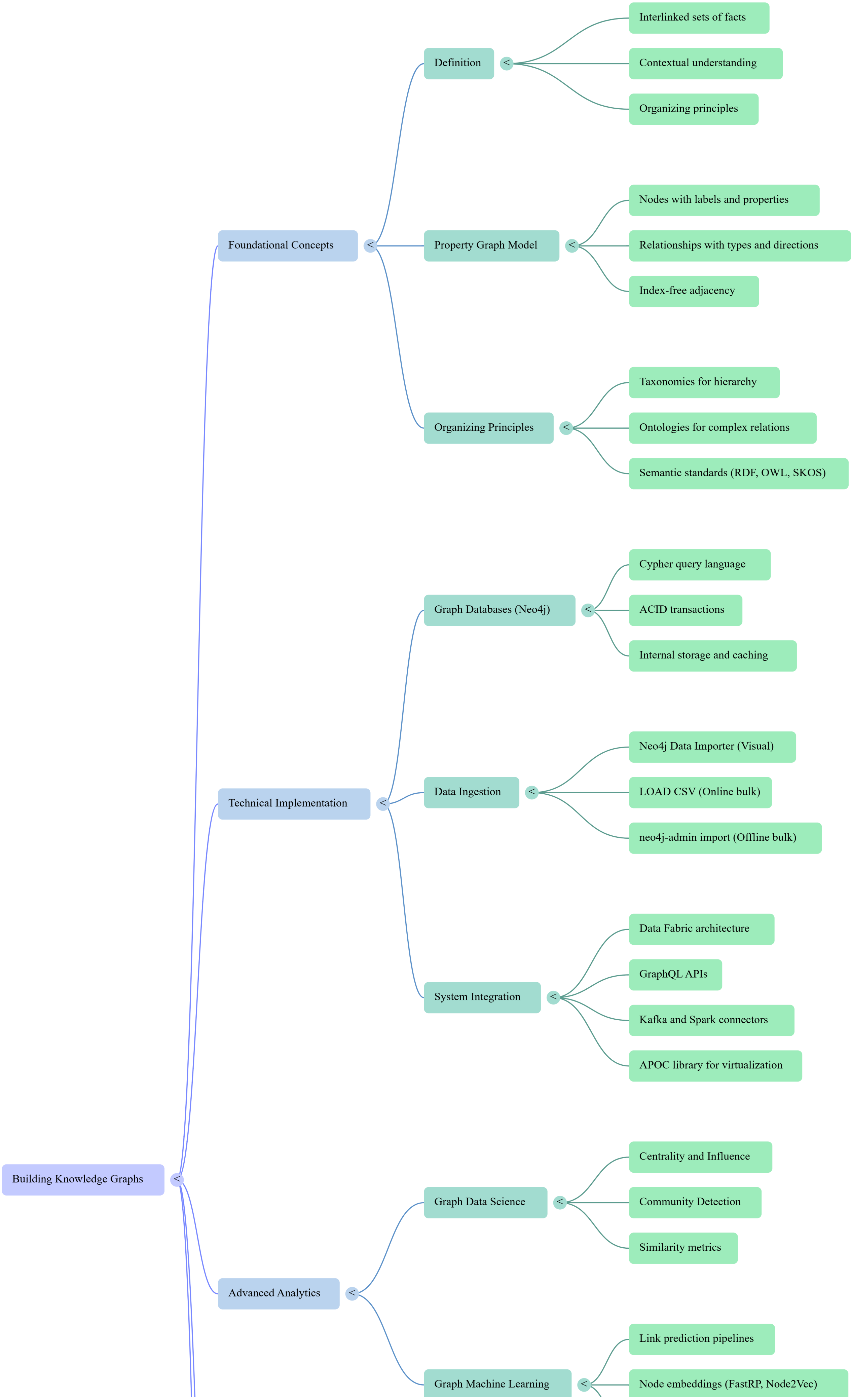

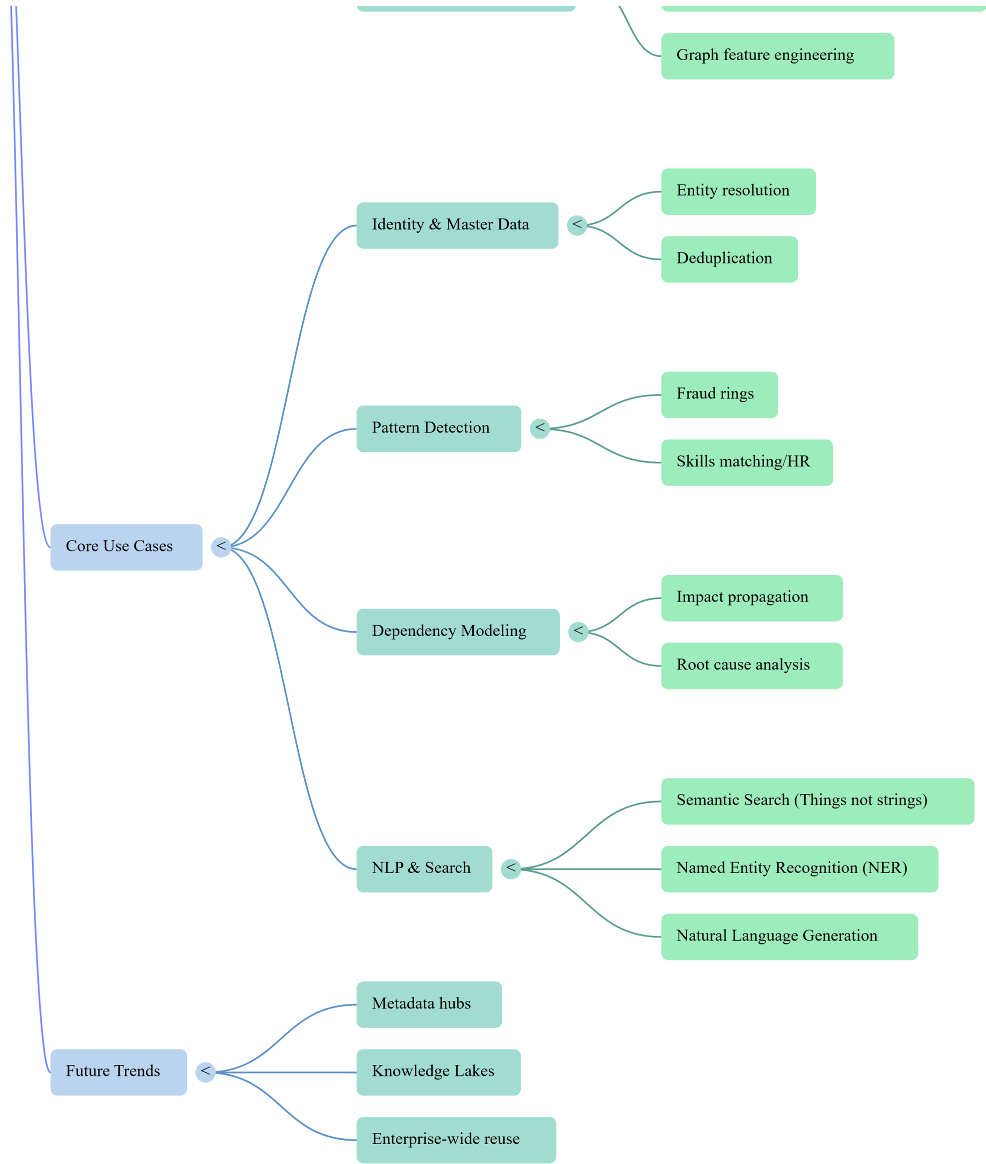

Flashcards easy

No	Question	Answer
1	What is the primary characteristic that transforms a plain graph into a knowledge graph?	The application of an organizing principle that provides contextual understanding.
2	In the property graph model, what do nodes represent?	Entities in the domain.

No	Question	Answer
3	What elements in a property graph represent how entities interrelate?	Relationships (or edges).
4	What are the key-value pairs stored on nodes or relationships called?	Properties.
5	True or False: In a property graph, relationships can exist without an end node.	False; relationships never dangle and must always have a start and an end node.
6	Which graph model element is used to declare a node's purpose, such as 'Customer' or 'Product'?	Labels.
7	Who invented graph theory in the 18th century to solve the Seven Bridges of Königsberg problem?	Leonhard Euler.
8	What is the difference between a graph and a chart?	Graphs show how things connect (networks), while charts visualize data distributions or functions (e.g., histograms).

No	Question	Answer
9	Why is the property graph model described as 'schema-ish'?	It is flexible and doesn't enforce a rigid schema but allows optional constraints on labels and properties.
10	What is the function of an 'organizing principle' in a knowledge graph?	It provides a layer of structure and context that allows humans and computers to reason about the data.
11	How does a taxonomy differ from a plain graph?	A taxonomy organizes categories into a broader-narrower or generalization-specialization hierarchy.
12	Which hierarchical relationship is typically used in a taxonomy to connect categories?	SUBCATEGORY_OF (or NARROWER_THAN/SUBCLASS_OF).
13	What is an ontology in the context of knowledge graphs?	A classification scheme that defines complex categories and varied relationship types beyond simple hierarchies.
14	Give an example of a horizontal (cross-cutting) relationship defined in an ontology.	PART_OF, COMPATIBLE_WITH, or DEPENDS_ON.

No	Question	Answer
15	What is the 'just-enough semantics' principle?	The practice of introducing only the semantic metadata required for current use cases to avoid over-engineering.
16	Name one standard language used to formally define ontologies.	Web Ontology Language (OWL) or RDF Schema.
17	What does SKOS stand for in the context of taxonomical classification?	Simple Knowledge Organization System.
18	Term: Cypher	Definition: A declarative, visual pattern-matching query language for graph databases.
19	In Cypher syntax, what symbols represent a node?	Parentheses ( ).
20	In Cypher syntax, what symbols represent a relationship?	Brackets within an arrow, such as -[ ]->.
21	Which Cypher keyword is used to insert data into a graph, even if it creates duplicates?	CREATE

No	Question	Answer
22	Which Cypher keyword acts as a mix of MATCH and CREATE, ensuring no duplicate patterns are created?	MERGE
23	What is the purpose of the DETACH DELETE command in Cypher?	It removes a node and all of its associated relationships simultaneously.
24	In graph database performance, query latency is typically proportional to what factor?	The amount of the graph traversed (not the total size of the database).
25	What is a 'graph local' query?	A query anchored to a specific, known node in the graph.
26	What is a 'graph global' query?	A query that processes the entire graph or large parts of it without being bound to a specific starting node.
27	Which Cypher keyword is used to visualize a query plan without executing it?	EXPLAIN

No	Question	Answer
28	Which Cypher keyword instruments and runs a query to provide runtime performance feedback?	PROFILE
29	Term: Index-free adjacency	Definition: A storage architecture where nodes and relationships point directly to each other via offsets, enabling $O(1)$ traversals.
30	How is the time complexity of accessing a node or relationship in Neo4j's traversal engine described?	Constant time, or $O(1)$.
31	In Neo4j, property storage is described as linear time, or _.	$O(N)$
32	What is the purpose of a 'write-ahead log' in graph databases?	To ensure ACID transactions and provide fault tolerance by persisting updates to disk before applying them.
33	Which tool provides a visual interface to map CSV files onto a graph domain model for loading into Neo4j?	Neo4j Data Importer

No	Question	Answer
34	Which Cypher clause transforms a list of values into individual rows for subsequent processing?	UNWIND
35	What is the primary command for importing data from web-based or local CSV files while the database is online?	LOAD CSV
36	For extremely large datasets (billions of records), which Neo4j command-line tool provides high-performance offline loading?	neo4j-admin import
37	In the context of information systems, what is a 'data fabric'?	A general-purpose data access layer that offers a unified, connected view of data across underlying silos.

No	Question	Answer
38	Which component of a data fabric often uses a knowledge graph to curate data across different storage types?	The enterprise index (or master data layer).
39	In Neo4j driver development, should you create one Driver object per query or one per application lifetime?	One per application lifetime (Driver objects are expensive).
40	Which Neo4j driver object is considered 'cheap' and intended for individual units of work?	Session
41	What is the purpose of 'parameterizing' queries in a database driver?	To increase performance by reusing query plans and to prevent injection attacks.
42	Which Neo4j feature allows multiple physical databases to be treated as a single unified virtual graph?	Composite Databases (Graph Federation).

No	Question	Answer
43	What is the APOC library in the Neo4j ecosystem?	A library containing hundreds of procedures and functions for common tasks like data loading and virtualization.
44	What is 'data virtualization' in a knowledge graph context?	Retrieving data from external sources on demand and presenting it as virtual nodes/relationships within the graph.
45	Which API framework is commonly used to expose knowledge graphs to frontend applications via a schema-based interface?	GraphQL
46	How can a knowledge graph integrate with real-time event streams like Apache Kafka?	Using a Kafka Connect plug-in to act as a source or sink for data events.
47	Which distributed data processing framework uses a specialized connector to treat Neo4j as a source or sink?	Apache Spark
48	What is the goal of graph data science?	To gain insight into a knowledge graph's structure using graph algorithms.

No	Question	Answer
49	Name the three broad categories of graph algorithms described in the text.	Statistical, Analytical, and Machine Learning.
50	What do 'Centrality' algorithms measure in a graph?	The influence or importance of individual nodes (e.g., bridges or bottlenecks).
51	Which algorithm class identifies tightly knit groups of nodes within a larger network?	Community Detection
52	What are the four phases of executing a graph algorithm in Neo4j Graph Data Science?	Read projection, Load projection, Execute algorithm, Store results.
53	What is a 'graph projection' in Neo4j Graph Data Science?	An in-memory, compressed representation of a specific subgraph optimized for parallel computation.
54	Define 'Betweenness Centrality'.	A metric representing the number of shortest paths that pass through a given node.

No	Question	Answer
55	In a production environment, why are 'secondary' servers often used for graph data science?	To isolate heavy computational workloads from the transactional processing of 'primary' servers.
56	What mechanism ensures a user sees their own writes even when querying an asynchronously updated secondary server?	Causal barrier (using Bookmarks).
57	In graph machine learning, what is 'in-graph machine learning'?	Using a knowledge graph to compute and predict its own missing elements, like relationships or labels.
58	What is 'graph feature engineering'?	Extracting topological features (like centrality or embeddings) from a graph to train external ML models.
59	What is a 'link prediction' algorithm used for?	Determining the likelihood that a relationship should exist between two nodes based on the existing topology.
60	Term: Node Embedding	Definition: An algorithmic encoding of a node's local topology into a fixed-length numerical vector.

No	Question	Answer
61	Which Neo4j algorithm is frequently used for high-performance node embeddings?	FastRP (Fast Random Projection).
62	What are the components of a link prediction pipeline in Neo4j GDS?	Node properties, link features, split configuration, model candidates, and training.
63	What does a 'metadata knowledge graph' map?	The location of data, the systems that process it (pipelines), and the final consumers (sinks).
64	In a metadata graph, what is a 'Data Sink'?	A terminal consumer of data, such as a dashboard visualization or an ML training set.
65	What is 'data lineage' in a metadata graph context?	Tracing the flow of data back to its original source platform across various tasks and datasets.
66	What is 'entity resolution'?	The process of determining if two separate records represent the same real-world thing.
67	What are the three fundamental activities in graph-based entity resolution?	Data preparation, Entity matching, and Master entity curation.

No	Question	Answer
68	In entity resolution, what is a 'blocking key'?	A value used to partition data into groups so that only likely matches are compared, reducing the search space.
69	What relationship type is used to connect nodes that definitively represent the same entity after resolution?	SAME_AS
70	What is the purpose of a 'SIMILAR' relationship in entity resolution?	To connect nodes that meet a probabilistic threshold for being the same entity, often annotated with a similarity score.
71	Which string similarity metric is mentioned in the text for matching weak identifiers like names?	Jaro–Winkler distance.
72	In the entity resolution process, what is a 'strong identifier'?	A unique, unambiguous value like a Social Security number or passport number.
73	Why is 'index-free adjacency' more efficient for graph traversals than using global join indexes?	It uses local pointers to move between records, avoiding the $O(\log N)$ cost of searching a global index for every step.

No	Question	Answer
74	In a Neo4j cluster, which server type is responsible for handling transactional writes?	Primary server.
75	Which GDS procedure mode writes results directly back to the nodes and relationships in the database?	write
76	Which GDS procedure mode streams results back to the caller without changing the graph stored in memory or on disk?	stream
77	Which GDS procedure mode enriches only the in-memory graph projection?	mutate
78	What is the primary benefit of treating metadata and data as separate layers within the same knowledge graph?	It allows users to search for data based on business concepts and provides built-in provenance and governance.

Flashcards medium

No	Question	Answer
1	What was Leonhard Euler's primary insight regarding the Königsberg bridge problem?	The physical geography was irrelevant; only the logical connections between landmasses mattered.
2	In the context of knowledge graphs, how do 'charts' differ from 'graphs'?	Charts visualize data trends (like histograms), while graphs model networks of entities and relationships.
3	What is the definition of a 'knowledge graph'?	An interlinked set of facts describing real-world entities and their relations in a machine-understandable format.
4	Why is an 'organizing principle' essential for a knowledge graph?	It provides a layer of structure and rules that allow humans or computer systems to reason about the underlying data.
5	In a Labeled Property Graph (LPG), what do 'nodes' represent?	Entities within a specific domain.
6	What are the three core elements that can be stored on a 'node' in a property graph?	Labels, properties (key-value pairs), and relationships.
7	What is the function of a 'label' in a property graph?	It declares the node's purpose or role in the graph, such as 'Customer' or 'Product'.
8	What rule governs 'relationships' in a property graph to ensure data integrity?	They never 'dangle'; they must always have a start node and an end node.

No	Question	Answer
9	In the LPG model, how are 'properties' defined?	They are key-value pairs representing entity data, such as a name or date of birth.
10	The property graph model is described as 'schema-ish' because _.	Constraints are applied only where needed rather than locking down the entire data model.
11	What distinguishes a 'plain old graph' from a 'knowledge graph'?	Plain old graphs encode their interpretation in application logic, while knowledge graphs include an explicit organizing principle.
12	Term: Taxonomy	Definition: A classification scheme that organizes categories in a broader-narrower or generalization-specialization hierarchy.
13	How does an 'ontology' differ from a 'taxonomy'?	Ontologies define complex, non-hierarchical relationships and semantics, such as 'PART_OF' or 'COMPATIBLE_WITH'.
14	What is the 'just-enough semantics' principle?	The practice of introducing semantic metadata only as current use cases demand to avoid overengineering.
15	Term: Semantic Bridge	Definition: An ontology that defines cross-equivalence to link the same concepts across different systems.
16	What is the primary design goal of the Cypher query language?	To be a visual, humane, and expressive declarative pattern-matching language.

No	Question	Answer
17	In Cypher syntax, what do parentheses `()` represent?	Nodes.
18	What is the purpose of the `MATCH` keyword in Cypher?	To compare user-defined patterns against the underlying knowledge graph data.
19	How does the `CREATE` keyword behave in Cypher compared to `MERGE`?	CREATE always inserts new data, while MERGE matches existing patterns or creates them if they are absent.
20	The Cypher keyword used to remove a node and all its attached relationships is _.	DETACH DELETE
21	What does a 'graph local query' mean?	A query anchored to a specific, known node in the graph rather than searching the entire database.
22	How does graph database size affect query latency in systems like Neo4j?	Latency is proportional to the amount of the graph traversed, not the total size of the database.
23	Term: Index-Free Adjacency	Definition: A storage technique where nodes store direct pointers to adjacent records, enabling $O(1)$ traversals.
24	What is the time complexity of reading a single property in Neo4j?	It is $O(N)$, where N is the number of properties on that specific node or relationship.
25	Why is 'scale-up' (adding RAM/CPU) generally preferred over 'scale-out' for graph processing?	Scale-up maintains data locality in memory, avoiding the high latency of network-based distributed traversals.

No	Question	Answer
26	What role does a 'write-ahead log' play in Neo4j internals?	It ensures durability by persisting intended updates to disk before they are applied to the graph.
27	What is the main advantage of the 'Neo4j Data Importer'?	It provides a no-code visual interface to map CSV files to a graph domain model.
28	When using `LOAD CSV` in Cypher, what does the `WITH HEADERS` clause do?	It allows the query to refer to CSV columns by their names rather than numerical indices.
29	Which Neo4j tool provides the fastest possible method for initial bulk data loading?	neo4j-admin import
30	Term: Data Fabric	Definition: A connected view of data across an organization that abstracts away the details of underlying third-party systems.
31	Regarding Neo4j drivers, why should a 'Driver' object be instantiated only once for an application's lifetime?	Driver objects are expensive to create because they manage authentication and network connection pools.
32	In a Neo4j cluster, what is the purpose of distinguishing between 'read' and 'write' transactions in application code?	It allows the driver to make better routing decisions to maximize cluster throughput.
33	What is 'Graph Federation'?	A technique using composite databases to provide a single unified entry point for multiple independent graph sources.

No	Question	Answer
34	How does 'Data Virtualization' with APOC benefit a knowledge graph?	It allows the graph to retrieve and present external data (like SQL or JSON) as virtual nodes without copying the data.
35	Term: Graph Data Science (GDS)	Definition: The application of graph algorithms to uncover structural insights, patterns, or features within a knowledge graph.
36	What are 'Statistical' graph algorithms primarily used for?	Providing high-level metrics like node counts, relationship types, and degree distributions.
37	What is the goal of 'Community Detection' algorithms?	To partition the graph into groups of nodes that are more densely connected to each other than to the rest of the network.
38	In GDS, what does 'Centrality' (or Influence) measure?	The relative importance or 'bridging' role of a node within the network topology.
39	What insight does the 'Betweenness Centrality' algorithm provide?	It identifies nodes that act as 'bridges' or bottlenecks by counting the shortest paths passing through them.
40	What is the first step in the Neo4j GDS workflow?	Creating a graph projection (loading a subgraph into memory for processing).
41	When executing a GDS algorithm, what does the 'mutate' mode do?	It writes the results of the algorithm back into the in-memory projection rather than the persistent database.

No	Question	Answer
42	How does 'In-graph Machine Learning' differ from traditional ML?	It allows a model to predict its own evolution (like missing links) by computing directly over its topology.
43	Term: Graph Feature Engineering	Definition: The process of extracting topological metrics (like PageRank or embeddings) from a graph to train ML models.
44	What is a 'Node Embedding' (such as FastRP)?	A numerical representation of a node's local topology, often used as a feature in machine learning.
45	In an ML Link Prediction pipeline, what is the purpose of the 'Split' phase?	To divide the graph into disjoint sets for training, testing, and validation.
46	What is the primary function of a 'Metadata Knowledge Graph'?	To act as an enterprise-wide map recording the location, shape, and provenance of data assets.
47	In a metadata graph, what does a 'Data Sink' represent?	A terminal consumer of data, such as a dashboard or an ML feature set.
48	Term: Data Lineage	Definition: The tracking of data origins, movements, and transformations across various systems and tasks.
49	What is 'Entity Resolution' (or Identity Resolution)?	The process of determining if two separate data records refer to the same real-world entity.
50	In identity graphs, why are 'Blocking Keys' used?	To reduce the search space by only comparing records that share a common high-level characteristic.

No	Question	Answer
51	What is the significance of a 'Golden Record' in master data management?	It is the single, trusted, and deduplicated 'master' view of an entity created from multiple sources.
52	In the entity resolution process, what is a 'SAME_AS' relationship used for?	To link nodes that have an unambiguous match on a strong identifier, like a Social Security number.
53	How does the 'Jaro-Winkler' distance assist in entity resolution?	It provides a string similarity metric used to compare weak identifiers like names.
54	What is the purpose of 'EXPLAIN' in Cypher development?	It allows a developer to visualize a query plan to understand how the database will execute the query without actually running it.
55	How does 'PROFILE' differ from 'EXPLAIN' in Cypher?	PROFILE actually executes the query and provides real-time instrumentation data on performance and database hits.
56	In GDS production, why are 'Secondary' server instances used?	To isolate heavy computational workloads from the transactional processing occurring on 'Primary' servers.
57	What does a 'Causal Barrier' (or Bookmark) ensure in a distributed Neo4j cluster?	It guarantees that a user will always see at least their own previous writes, even when querying different servers.
58	In a metadata graph, what is the 'Task' entity used to model?	Any discrete data job that processes or transforms a data asset.

No	Question	Answer
59	Which algorithm class would you use to predict if two users in a social network should be friends?	Link Prediction.
60	The property of 'ACID' transactions that ensures all parts of a transaction succeed or none do is _.	Atomicity
61	Term: Raft	Definition: The consensus algorithm used by Neo4j to maintain synchronized transaction logs across a cluster.
62	Why is property access in graphs considered linear time ($O(N)$)?	Properties are stored as linked lists, requiring the database to scan the list to find a specific key.
63	In the Cypher `UNWIND` clause, what is the result of applying it to a list?	It transforms a single list of values into individual rows for subsequent query steps.
64	What is the 'Cold Start' problem in recommendation systems?	The challenge of providing recommendations when there is little or no historical interaction data for a user or item.
65	What is the function of the `WHERE` clause in Cypher?	It adds logical predicates to a `MATCH` pattern to filter results based on properties or relationships.
66	A relationship that goes from one node back to the same node is called a _.	Self-relationship (or loop)

No	Question	Answer
67	In the context of the 'Data Importer', what is the role of an 'ID column'?	It serves as a temporary unique identifier used to link nodes and relationships during the bulk import process.
68	What does 'Index-Free Adjacency' eliminate the need for during a traversal?	Global join indices or recursive lookups.
69	Term: Pathfinding Algorithms	Definition: A class of algorithms used to find the most efficient route (shortest, fastest, cheapest) between nodes.
70	In an Identity Knowledge Graph, what is 'Data Preparation'?	The phase where heterogeneous data is cleansed, standardized, and harmonized to make fields comparable.
71	In Neo4j, what happens to a relationship if one of its endpoint nodes is deleted?	The database prevents the deletion unless the relationship is also explicitly deleted, as relationships cannot dangle.
72	How does 'Dijkstra’s Algorithm' calculate the shortest path?	It explores the graph by evaluating the cumulative weights of relationship properties (like distance).
73	What is the primary benefit of using 'GraphQL' with a knowledge graph?	It provides a strictly typed API schema that allows clients to request exactly the graph data they need.
74	In GDS, what is the purpose of the 'Graph Catalog'?	It manages and stores named in-memory graph projections for reuse across different algorithmic tasks.

No	Question	Answer
75	What is the risk of an 'Overly Ambitious Metamodel' early in a project?	It leads to expensive, slow efforts that may become out-of-date before providing business value.
76	Term: Hypergraph	Definition: A graph model where a single relationship (edge) can connect more than two nodes.
77	In Cypher, what is the difference between `SET` and `REMOVE`?	SET adds or updates properties and labels, while REMOVE deletes them without deleting the node itself.
78	How can 'Weak Identifiers' be used to link entities safely?	By combining them into similarity scores and applying correcting factors based on other non-identifying features.
79	Why is 'Context' considered the defining feature of a knowledge graph?	Context is provided by the associative network of relationships and metadata that defines how entities interrelate.
80	What occurs during the 'Execute' phase of the GDS workflow?	The chosen graph algorithm runs against the loaded in-memory projection to compute scores or patterns.

Flashcards hard

No	Question	Answer
1	Knowledge Graph Definition	Interlinked sets of facts describing real-world entities and their interrelations in a human- and machine-understandable format with an organizing principle.

No	Question	Answer
2	In the context of knowledge graphs, what is the primary role of an 'organizing principle'?	It provides a layer of structure and context that enables reasoning and discovery over the underlying data.
3	How does the 'organizing principle' function as a contract between a knowledge graph and its users?	It allows consuming software to expect specific structures, like labeled nodes and typed relationships, without needing detailed domain knowledge.
4	Term: Labeled Property Graph Model	A graph model consisting of nodes, labels, relationships, and properties used to build high-fidelity domain representations.
5	In the property graph model, what primitive is used to represent an entity?	A node.
6	Within a property graph, what do 'labels' signify on a node?	The node's specific role or category within the graph domain.
7	What is the rule regarding 'dangling' relationships in a property graph model?	Relationships must always connect a start node and an end node.
8	In a property graph, where can 'properties' (key-value pairs) be stored?	Properties can be stored on both nodes and relationships.
9	How does 'index-free adjacency' allow for constant time ($O(1)$) traversals?	It uses physical offsets (pointer-chasing) to move between nodes rather than global indexes.

No	Question	Answer
10	Term: Taxonomy	A classification scheme that organizes categories into a broader-narrower or generalization-specialization hierarchy.
11	What is the primary structural difference between a taxonomy and an ontology?	Ontologies support complex, multi-level relationships and horizontal associativity beyond simple hierarchies.
12	The principle of 'just-enough semantics' advocates for what development approach?	Implementing semantic metadata incrementally based on current use cases rather than building an all-encompassing model upfront.
13	In graph database architecture, what is the 'causal barrier'?	A mechanism ensuring that clients always see at least their own writes across a distributed cluster.
14	Which Cypher keyword is used to ensure a pattern exists without creating duplicates?	MERGE
15	What happens if a MERGE statement only partially matches an existing pattern in the graph?	The database fails to match the pattern and creates a new instance of the entire pattern, potentially resulting in duplicates.
16	To remove a node along with all its connected relationships in Cypher, use the _ command.	DETACH DELETE
17	What is the primary design goal of the Cypher query language?	To be a humane, expressive, and predominantly visual pattern-matching language.

No	Question	Answer
18	In Cypher syntax, what do parentheses `()` represent?	Nodes
19	How is a directed relationship with a type specified in Cypher ASCII art?	-[:TYPE]->
20	Concept: Graph Local Query	A query anchored to specific known nodes, exploring the immediate or deep neighborhood of those nodes.
21	What distinguishes a 'Graph Global Query' from a local one?	Global queries process large parts or the entirety of the knowledge graph to generate aggregates without being bound to a specific starting node.
22	Cypher syntax: `[:FRIEND*1..2]`	Matches a relationship of type FRIEND at a path depth between one and two hops.
23	Why is 'parameterization' essential when sending queries via a database driver?	It allows the database to reuse query plans and protects the system against injection attacks.
24	What is the purpose of the Cypher `EXPLAIN` keyword?	It visualizes the query plan to help developers minimize 'DB hits' without executing the query.
25	How does `PROFILE` differ from `EXPLAIN` in Cypher performance tuning?	PROFILE executes the query and instruments it to show actual runtime behavior and execution costs.

No	Question	Answer
26	In Neo4j internals, where are the 'neighbors' of a node essentially stored?	They are stored in a relationship chain directly referenced by the node, acting as a local index.
27	Why are property stores in Neo4j considered linear time ($O(N)$) for reads?	Retrieving a specific property requires iterating through a list of property records attached to the node or relationship.
28	What role does the 'write-ahead log' play in Neo4j's ACID transactions?	It ensures durability by persisting intended updates to disk in an ordered fashion before applying them to the graph.
29	The 'Raft' algorithm is utilized in Neo4j clusters to achieve what purpose?	To maintain transaction log consistency and synchronization across multiple servers.
30	Which bulk-loading tool is best for high-speed, offline initial bootstrapping of billions of records?	neo4j-admin import
31	What is the function of the `UNWIND` clause in Cypher ingestion scripts?	It transforms a list of values into individual rows to be processed by subsequent statements.
32	In the `LOAD CSV` command, what is the purpose of the `WITH HEADERS` sub-clause?	It allows the query to reference data columns by their header names rather than by numerical index.
33	Term: Data Fabric	An organization-wide data access layer providing a connected, curated view of data across disparate underlying systems.

No	Question	Answer
34	What is a 'Composite Database' in the context of graph federation?	A virtual database that provides a single point of access to multiple physical graph data sources.
35	How can 'virtual nodes' be created from external SQL or JSON sources without copying data?	By using procedures from the APOC library to map external results into the Cypher query context.
36	What does the `@relationship` directive do in Neo4j's GraphQL implementation?	It maps GraphQL fields to relationships between nodes in the underlying graph database.
37	Category: Influence Algorithms (GDS)	Algorithms that measure a node's importance or 'centrality' based on its position and connections in the network.
38	What does 'Betweenness Centrality' calculate?	The number of shortest paths that pass through a specific node, identifying bridges and bottlenecks.
39	In Graph Data Science (GDS), what is a 'graph projection'?	An in-memory, compressed representation of a specific subgraph optimized for high-performance algorithm execution.
40	What is the operational difference between the GDS `mutate` and `write` modes?	Mutate saves results back into the in-memory projection, while Write persists them as properties in the underlying database.

No	Question	Answer
41	Concept: Community Detection	Algorithms that partition a graph into sets of nodes that are more densely connected to each other than to the rest of the network.
42	In GDS, what is 'Topological Machine Learning'?	Using the graph's structure and connectivity patterns as the primary features for predictive modeling.
43	Term: Node Embedding	A numerical vector that encodes a node's local topology and relative position within the graph.
44	How does 'Graph Feature Engineering' improve machine learning models?	It extracts higher-quality features derived from graph topology, such as PageRank or community IDs, to train predictive models.
45	In a GDS link prediction pipeline, what is the role of the 'combiner' (e.g., Cosine or Hadamard)?	It merges individual node properties into a single feature vector representing the potential relationship between a pair of nodes.
46	Term: Metadata Knowledge Graph	An enterprise map recording the location, schema, and provenance of data assets, processing tasks, and consumers.
47	What is a 'Data Sink' in a metadata graph model?	A terminal consumer of data, such as a dashboard or ML feature set, that does not produce a new dataset.

No	Question	Answer
48	Term: Entity Resolution	The process of determining whether two records from different data sources refer to the same real-world entity.
49	What is the purpose of 'Blocking Keys' in entity resolution?	To reduce the search space by only comparing records that share a common attribute, preventing n^2 complexity.
50	In identity matching, how does a 'Strong Identifier' differ from a 'Weak' one?	Strong identifiers (like SSNs) provide unambiguous matches, while weak identifiers (like names) require probabilistic or fuzzy scoring.
51	What is the 'Jaro-Winkler' metric used for in data preparation?	Measuring the similarity between two strings to identify potential duplicates in fuzzy matching rules.
52	Concept: Golden Record	A single, trusted master entity created by merging multiple resolved duplicate records from various source systems.
53	How does the 'FastRP' algorithm function in a GDS pipeline?	It provides a fast method for generating node embeddings by projecting the graph's topology into a lower-dimensional vector space.
54	In a metadata graph, what does 'Data Lineage' analysis track?	The flow of data from its origin platforms through various transformation tasks to its final consumption points.
55	What logic does step 3 of the 'Graph-Based Entity Resolution' process apply to SIMILAR relationships?	It discards SIMILAR links if there is a known mismatch in strong identifier features like Passport Number or SSN.

No	Question	Answer
56	Why is relationship 'direction' mandatory when storing data using Cypher?	It ensures a high-fidelity model by disambiguating the semantic relationship between nodes.
57	What is the primary benefit of using 'Secondary Servers' in a Neo4j cluster for GDS?	It provides physical isolation, preventing compute-intensive analytical workloads from impacting transactional performance.
58	In GDS, what is the 'Alpha' tier for procedures?	Experimental procedures that may change without notice and are not yet recommended for production use.
59	How does 'Link Prediction' differ from 'Node Classification' in graph machine learning?	Link prediction forecasts missing connections between nodes, whereas node classification predicts the label or category of a node.
60	Term: Semantic Search	A search technique that uses an organizing principle to find relevant information based on meaning rather than literal string matching.
61	In GDS pipeline splitting, what is the 'Feature Set'?	The portion of the graph reserved to compute the properties used as features for the machine learning model.
62	What is the function of the `gds.find_node_id` helper in the GDS Python API?	It retrieves the internal database ID for a node based on its labels and property values.

No	Question	Answer
63	Within a metadata KG, what relationship typically connects a 'Dataset' to a 'DataPlatform'?	The source relationship.
64	Term: Data Stewardship	The management and oversight of an organization's data assets to ensure accessibility, quality, and compliance.
65	How can 'Betweenness Centrality' identify critical infrastructure in a transport network?	By highlighting hubs where the failure of a single node would most significantly disrupt overall network connectivity.
66	What is 'Causal Consistency' in a graph database?	A consistency model that ensures the order of related operations is preserved across all nodes in a cluster.
67	In Cypher, what does the keyword `YIELD` signify when calling a procedure?	It selects specific output columns from the procedure results to be used in the remainder of the query.
68	What is the role of a 'Driver' object in application-to-graph database connectivity?	It manages the connection pool and protocol details for communicating with the database server.
69	Cypher: `MATCH (n) DETACH DELETE n`	Effectively deletes the entire graph by removing all nodes and their associated relationships.
70	How does 'Graph Federation' via Neo4j Fabric handle cross-database queries?	It allows a single Cypher query to target multiple graphs by using the `USE` clause to specify different database aliases.

No	Question	Answer
71	Term: Semantic Bridge	An ontology used to link and map concepts between different systems to a standard, well-understood vocabulary.
72	In the context of GDS, what does 'Scale-Up' vs 'Scale-Out' imply for graph processing?	Scale-up uses a single large-memory machine for locality benefits, whereas scale-out distributes data across servers at the cost of network latency.
73	Why is 'Harmonization' a critical step in the Identity KG data preparation phase?	It aligns disparate data formats (e.g., age vs. date of birth) into a common representation to make them comparable.
74	What does a `SAME_AS` relationship indicate in an identity knowledge graph?	An unambiguous, high-confidence link between two records based on a shared strong identifier.
75	In GDS `train` results, what does 'AUCPR' measure?	The Area Under the Precision-Recall curve, used to evaluate the quality of a predictive model.
76	How does the 'Louvain' algorithm identify communities?	It optimizes modularity by iteratively grouping nodes to maximize the density of internal connections versus external ones.
77	What is the primary risk of 'Boiling the Ocean' in knowledge graph modeling?	Wasting time and resources capturing an overly ambitious, monolithic metamodel that may be out of date before completion.

No	Question	Answer
78	Cypher: `RETURN count{ (d)<-[:consumes]-(:DataSink) }`	An inline subquery used to count the number of DataSink nodes that have an incoming 'consumes' relationship from node d.
79	In GDS, what is the purpose of the 'Graph Catalog'?	It manages named graph projections in memory so they can be reused across multiple algorithm executions.
80	What is 'Index-Free Adjacency'?	A storage technique where nodes directly store the memory addresses of their adjacent relationships, bypassing the need for global lookups.

Source: